CONNEXUS black sea tech forum | 2026

What If Energy Could Think?
AI From Forecasting to Autonomous Energy Decisions



Iliya Zheliazkov
Software Engineer, T&D Engineering

ICT CLUSTER VARNA

BSMEPA
BULGARIAN SMALL AND MEDIUM ENTERPRISES PROMOTION AGENCY

BAPHA

BRAIN++ AI FACTORY

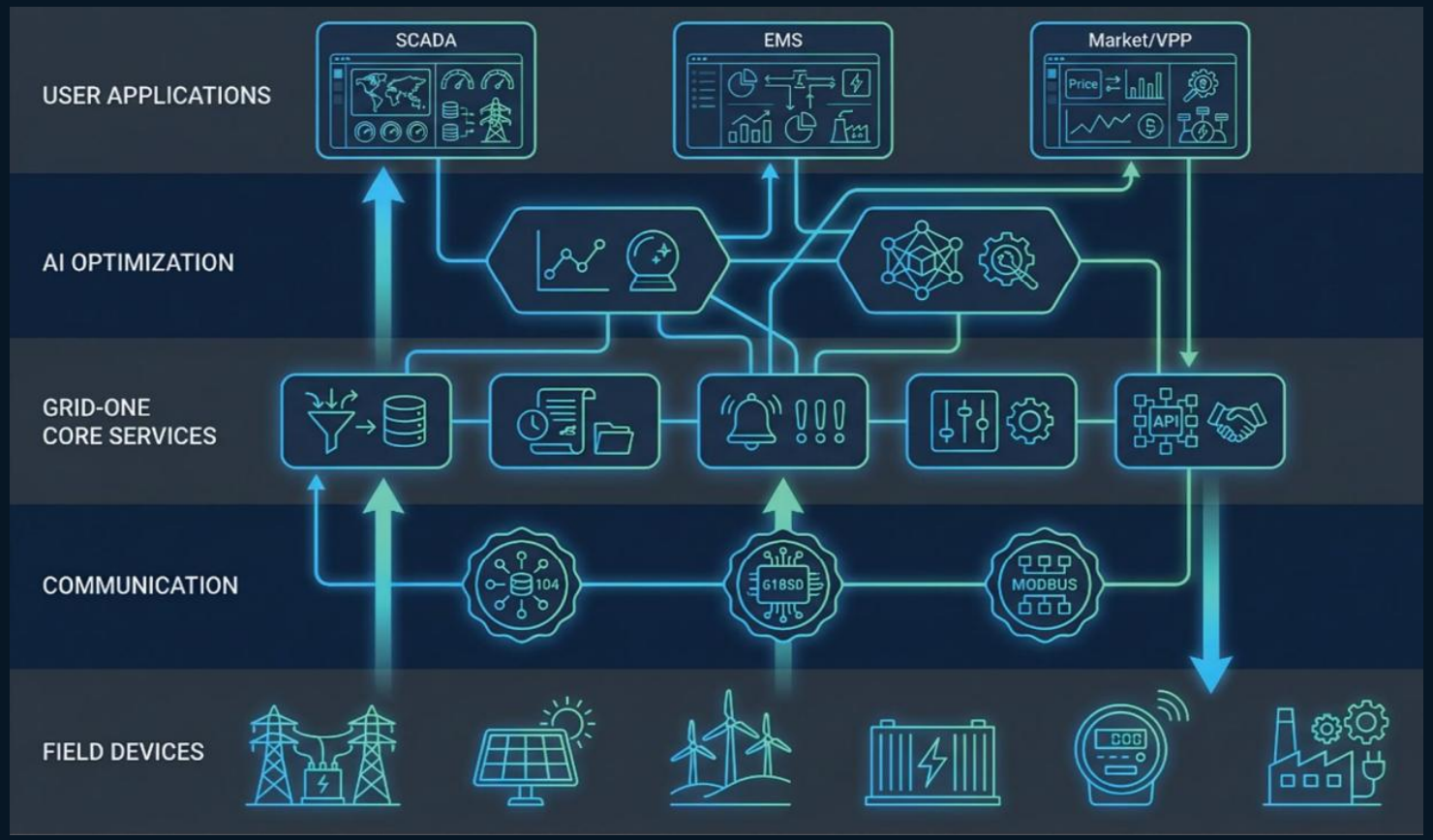
ARC FUND

Grid-One

AI for the Energy System

What If Energy Could Think?

AI From Forecasting to Autonomous Energy Decisions



FIELD DEVICES → COMMUNICATION → CORE SERVICES → AI OPTIMIZATION → SCADA / EMS / MARKET

Working AI across the whole operational stack — not one isolated model.

The Last 12 Months Compressed Years of Change

For us, AI did not suddenly appear. The operating problem accelerated — especially around large-scale storage and integration.

BULGARIA — BESS MOVED FROM EMERGING TO INDUSTRIAL SCALE

BESS SCALE	VENDOR DIVERSITY	DOCS IN FLUX	ENGINEERING ATTENTION	NEW CONSTRAINTS
multi-container systems more hierarchy	interfaces • protocols local tools • language	manuals • firmware parameters	commissioning diagnostics	market • grid • lifecycle environment

TWO PRACTICAL AI RESPONSES

AI FOR THE ENERGY SYSTEM

forecast • plan • schedule • optimize • detect risk

AI FOR THE ENGINEERING TEAM

docs / RAG • translation • site creation • configuration • diagnosis

Operational complexity started scaling faster than engineering capacity.

The Energy System Has Changed

From a mostly one-directional system to a distributed, bidirectional decision environment.

THEN

Centralized energy flow

GENERATION

↓

GRID

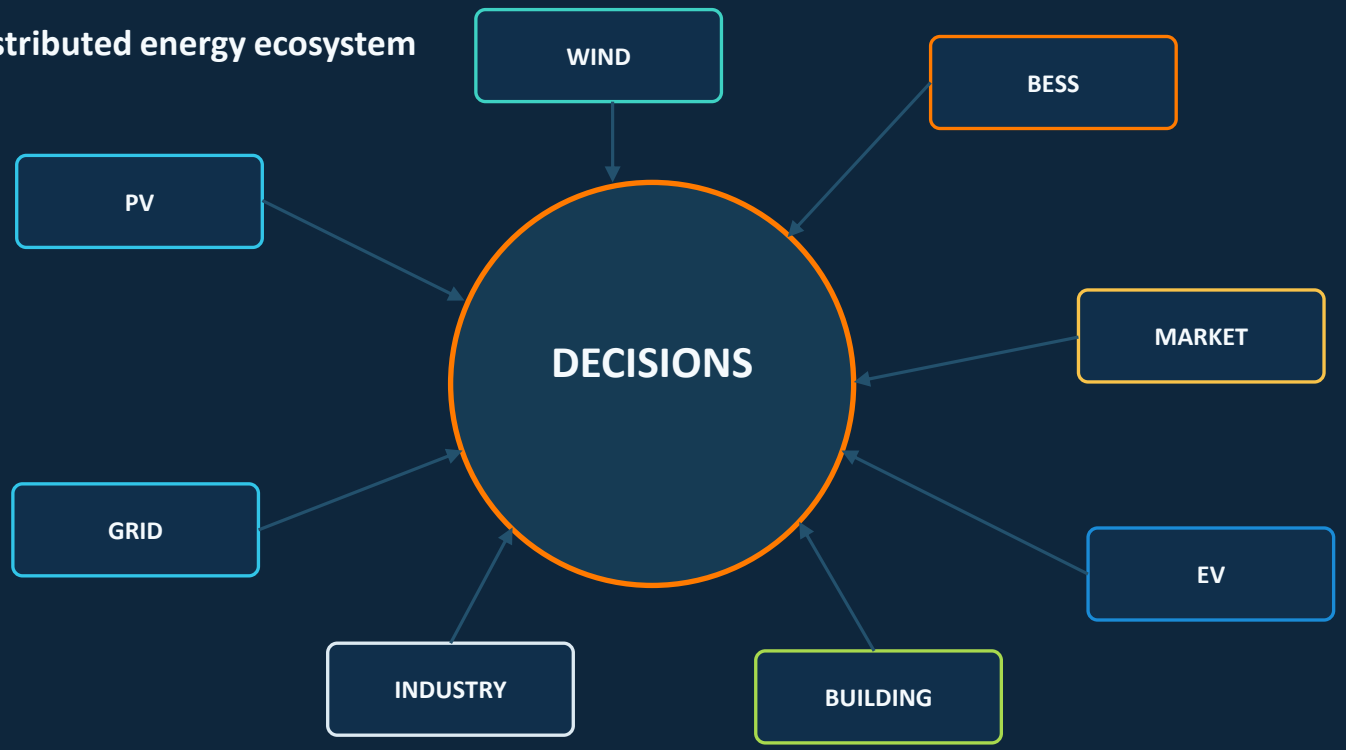
↓

CONSUMER

Few actors
Mostly predictable direction
Limited controllability

NOW

Distributed energy ecosystem

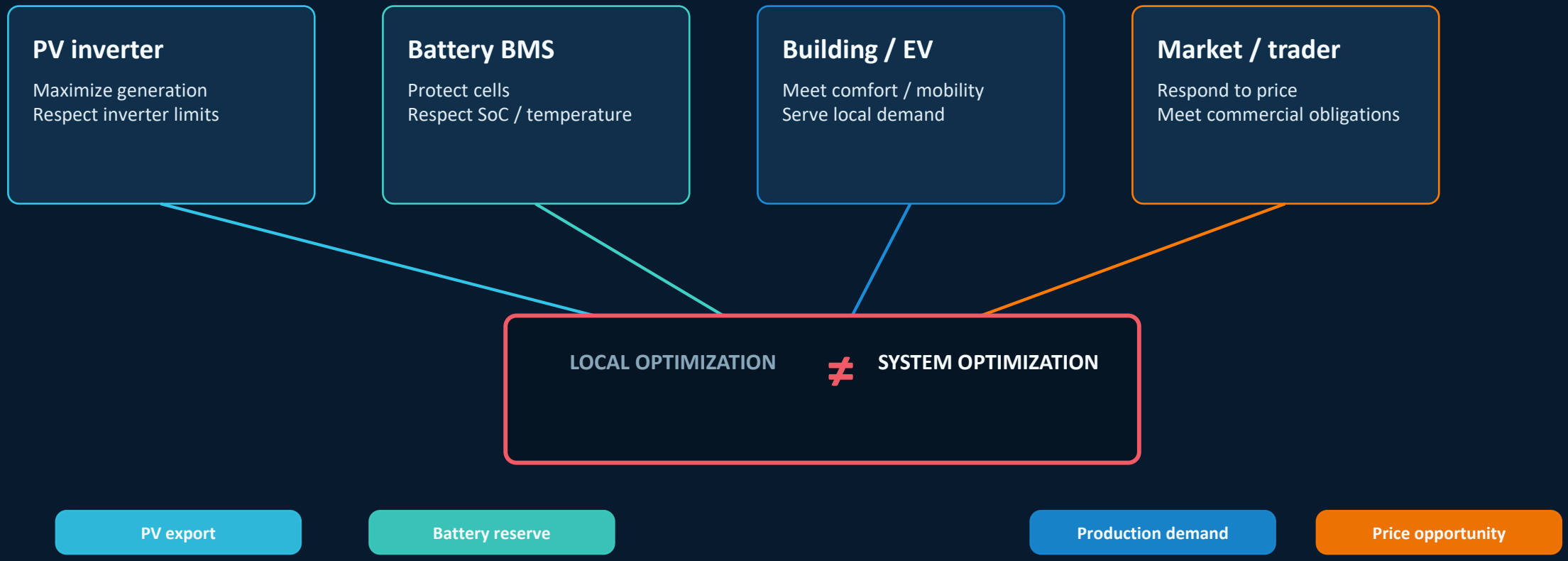


The number of interdependent decisions has exploded.

milliseconds seconds minutes hours day-ahead lifecycle

Smart Assets ≠ Smart Energy System

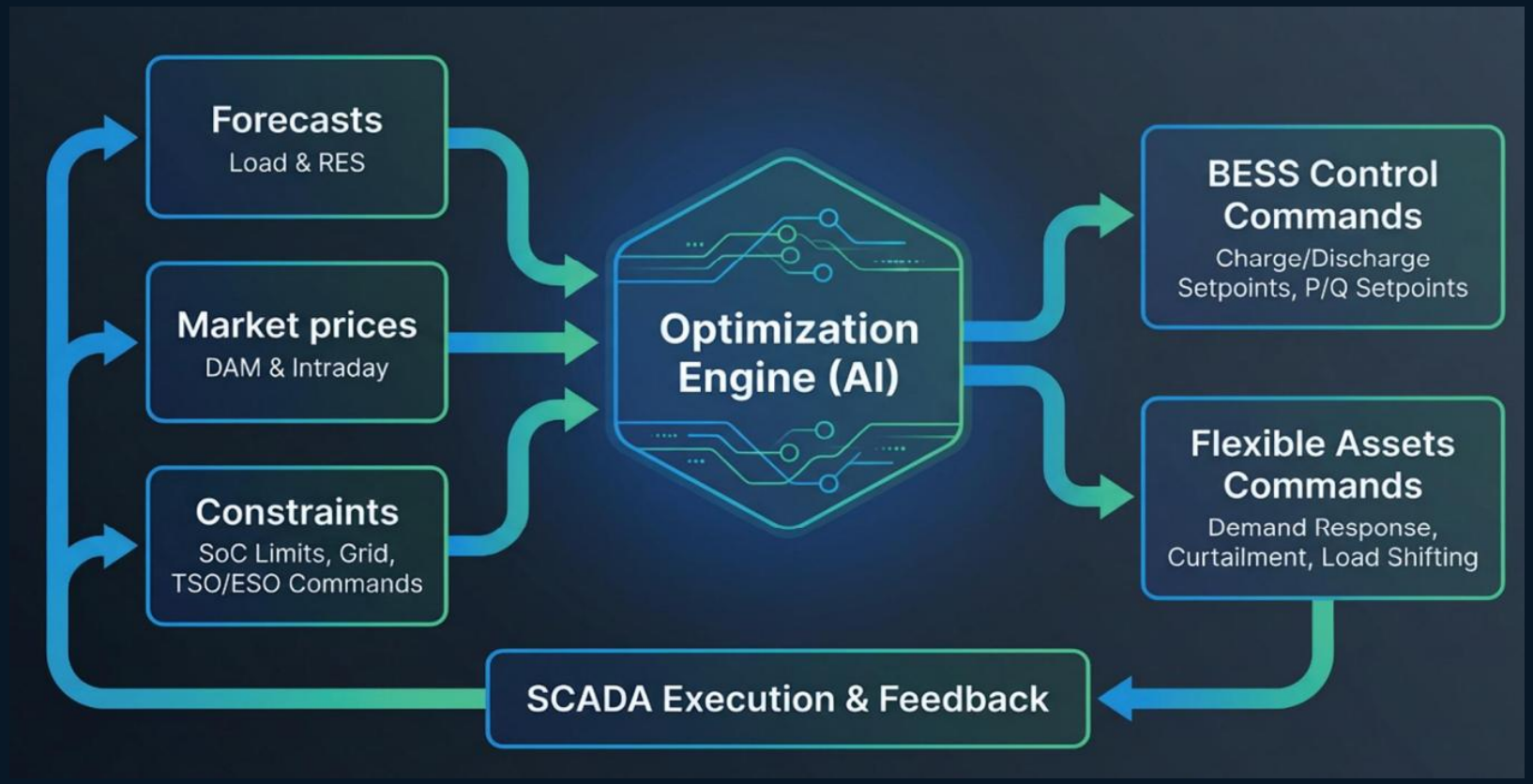
Local intelligence — and vendor-specific management software — can still leave system-level decisions fragmented.



The missing layer is coordinated intelligence across assets, constraints and objectives.

What Does It Mean for Energy to Think?

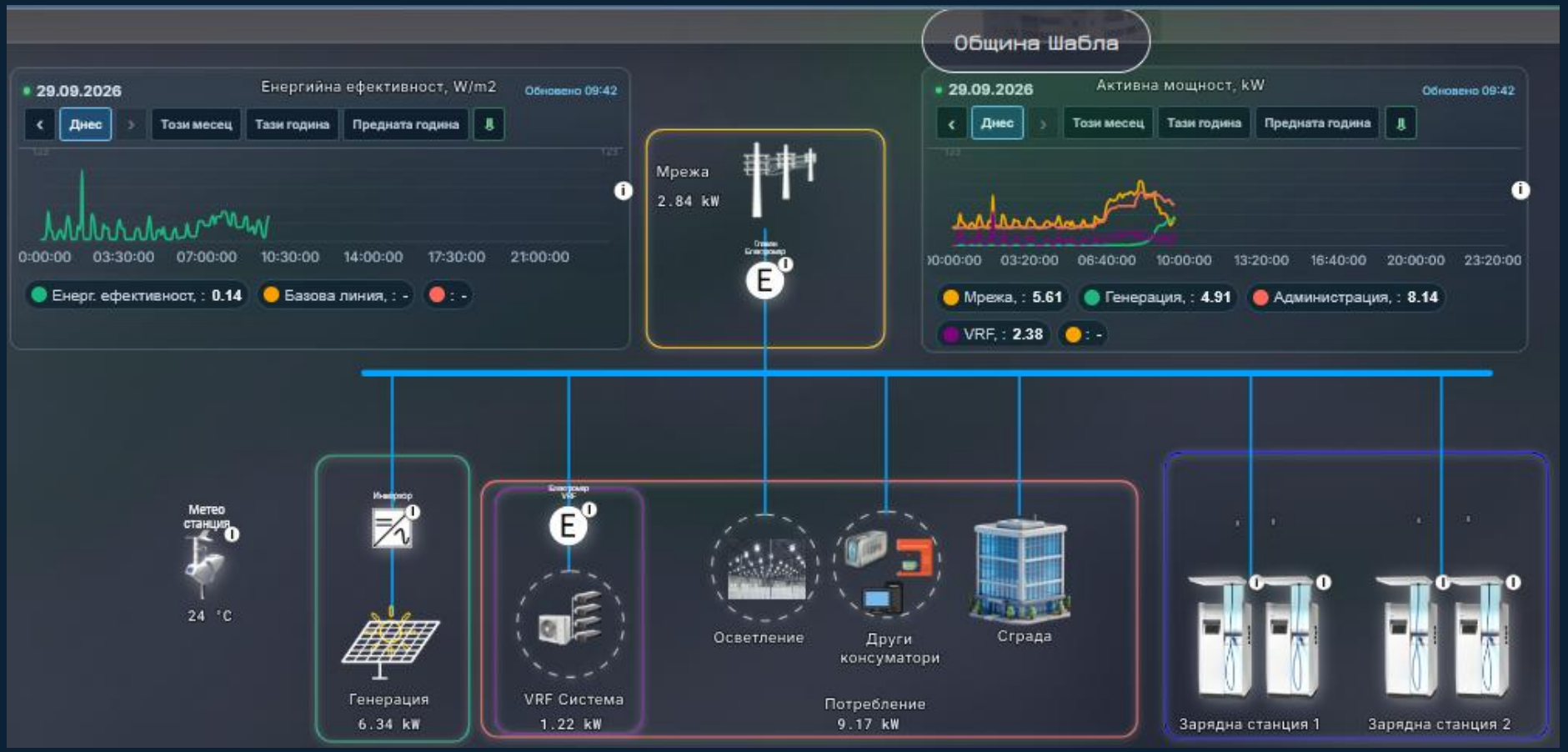
Prediction creates value only when it changes a decision — and that decision can be executed safely.



Forecasts + markets + constraints → optimization → commands → SCADA execution & feedback.

Shabla Municipality — Energy and Building Systems

PV • building loads • VRF • EV charging • weather • grid

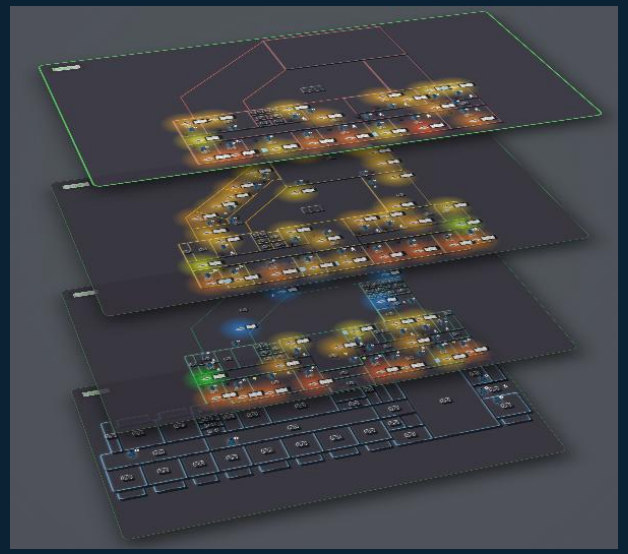


One operational view across energy production, consumption, building systems and e-mobility.

Shabla Municipality — Building AI from Equipment to Whole Site

One model for each intelligent equipment type — plus one model for the overall site.

FOUR-FLOOR OPERATING CONTEXT



EQUIPMENT MODELS

HVAC • lighting • room conditions
individual intelligent equipment

WHOLE-SITE MODEL

coordinates site-level behaviour

REAL BUILDING SIGNALS



occupancy • temperature • humidity • illuminance

scheduled HVAC • lighting control • manual override

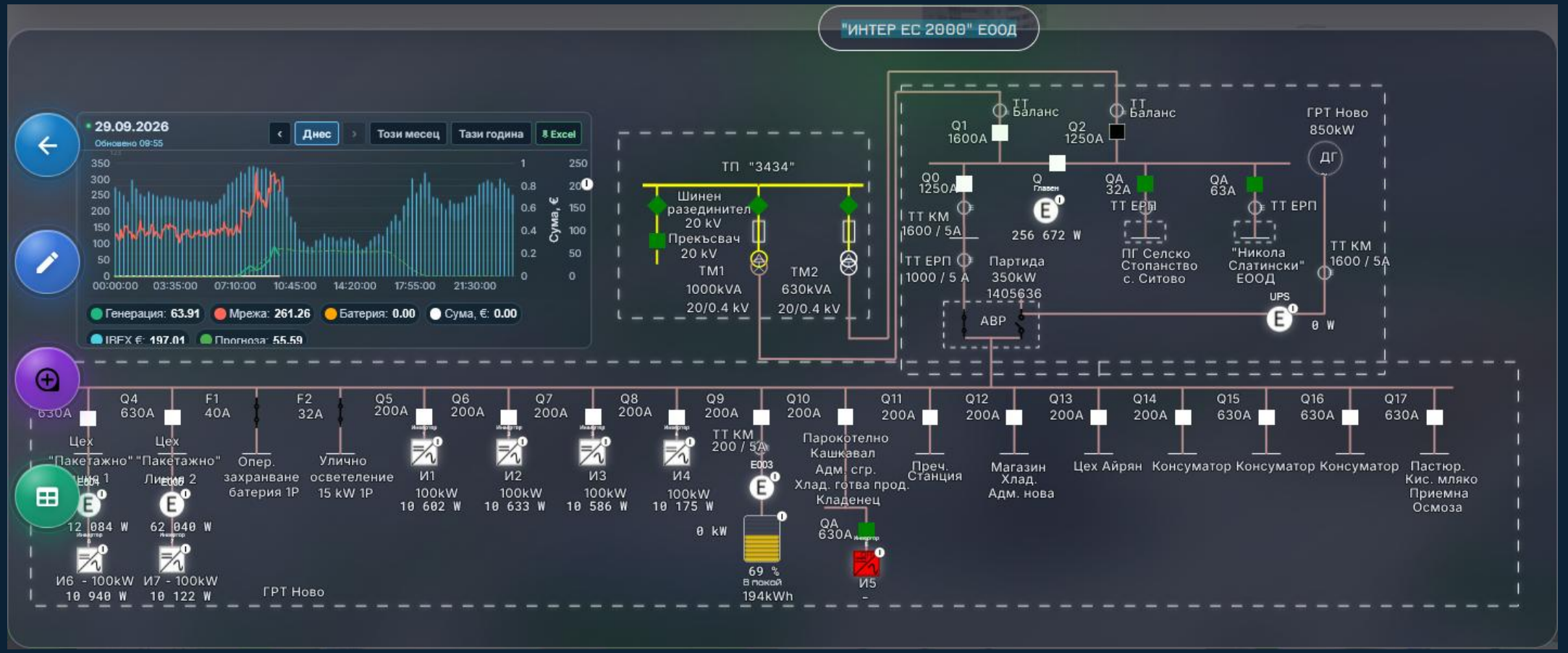
PARAMETERIZED TRAINING TEMPLATE

reuse the learning structure, then parameterize it for each building/site

Equipment intelligence becomes useful when the overall site can coordinate it.

INTER ES 2000 — Industrial Energy in One Operational View

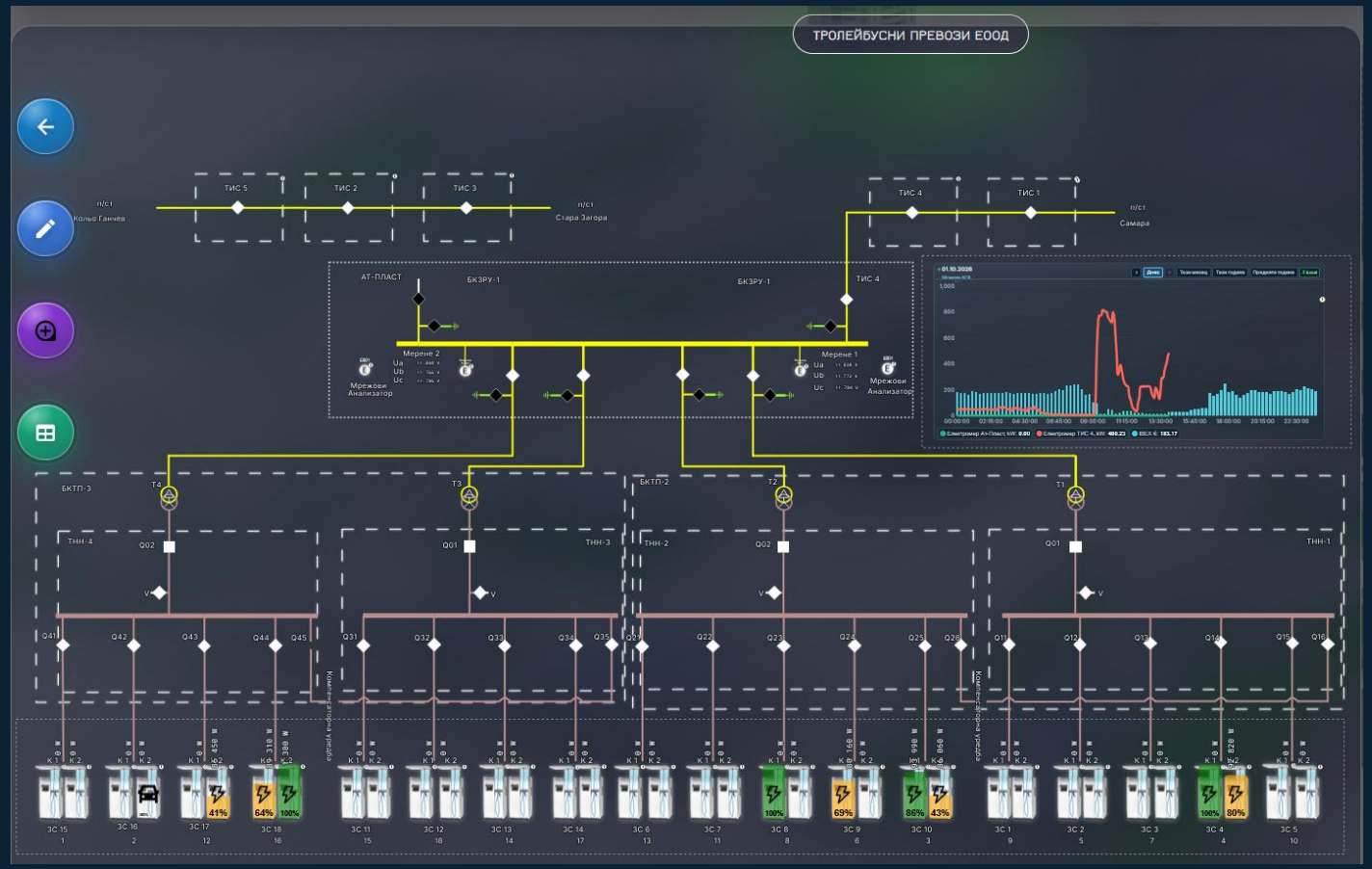
Electrical distribution • transformers • PV • BESS • production loads



Generation, storage and industrial demand share the same electrical constraints.

Trolleybus Transport — Charging Around the Service Schedule

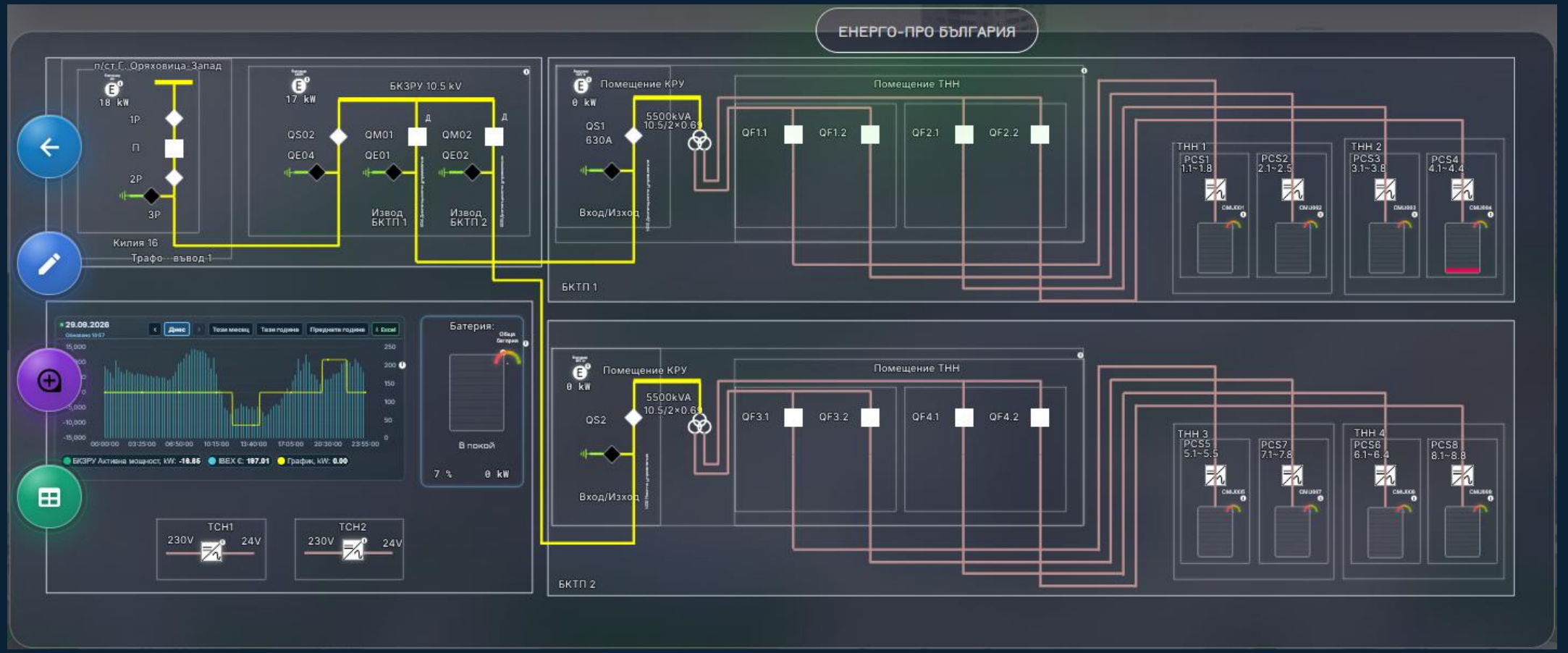
Shift charging toward lower-cost periods while keeping buses ready for scheduled service.



Charging time becomes a controllable cost — within the transport schedule.

ENERGO-PRO Bulgaria — BESS in MV Infrastructure

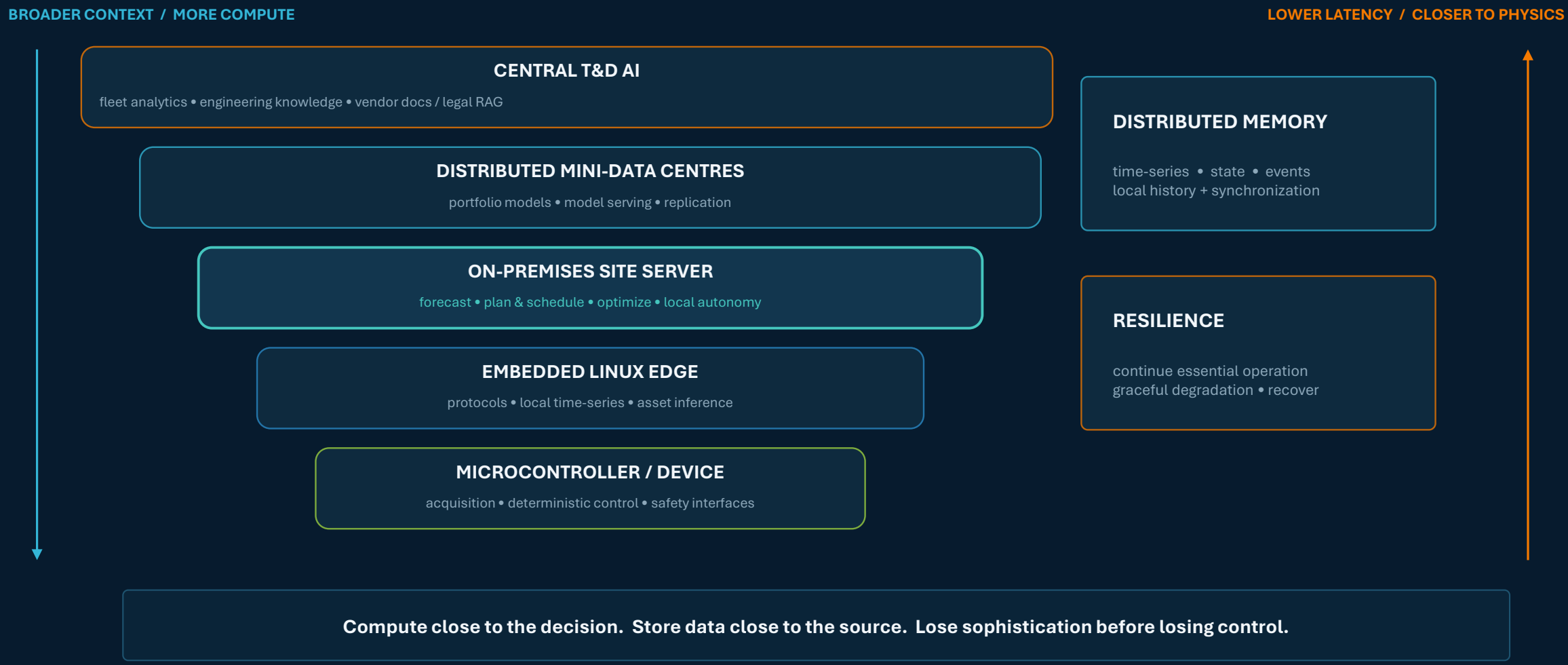
MV topology • transformers • PCS • standard BESS charge/discharge



Reliable charge/discharge control starts with the real MV topology beneath the battery.

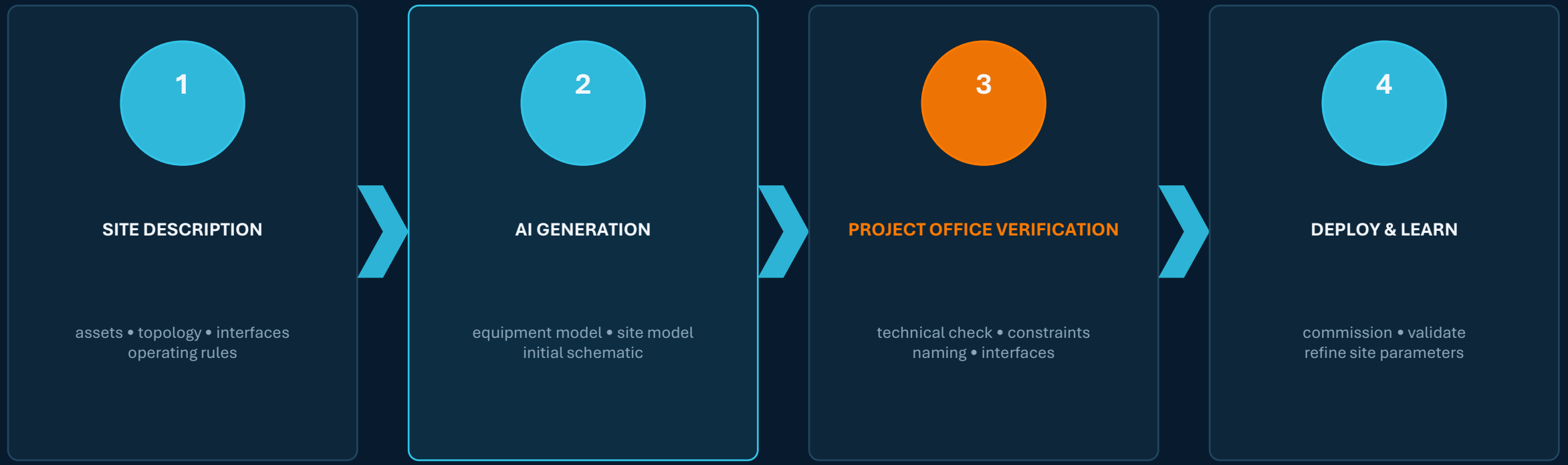
Intelligence Where It Belongs

Put intelligence where latency, compute and resilience require it — without giving up local control.



AI-Assisted Site Creation

A site description can become an operational model and schematic faster — with technical verification before deployment.

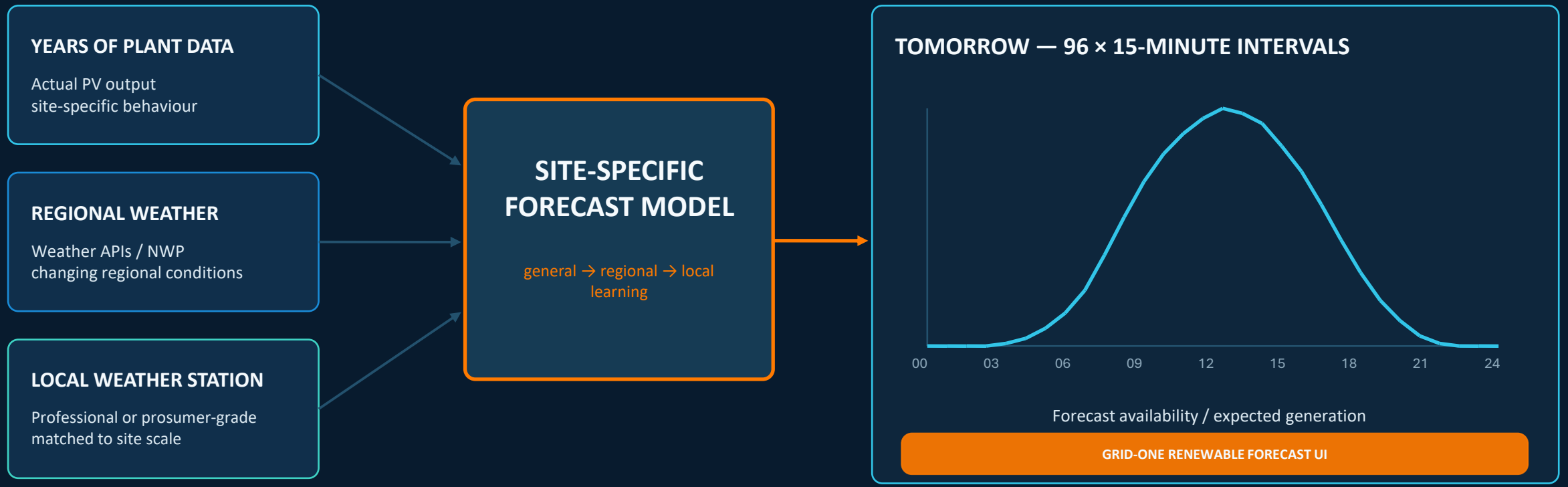


AI accelerates engineering preparation. Technical responsibility remains with the project office.

Scale the engineering workflow without removing engineering verification.

Prediction Becomes Operational

Site-specific renewable forecasting turns weather and history into a schedule-ready energy forecast.



COLD START → **LEARN FROM REGION** → **LEARN FROM SITE** → **MATURE SITE MODEL**

Industry / physical models

Nearby / similar sites

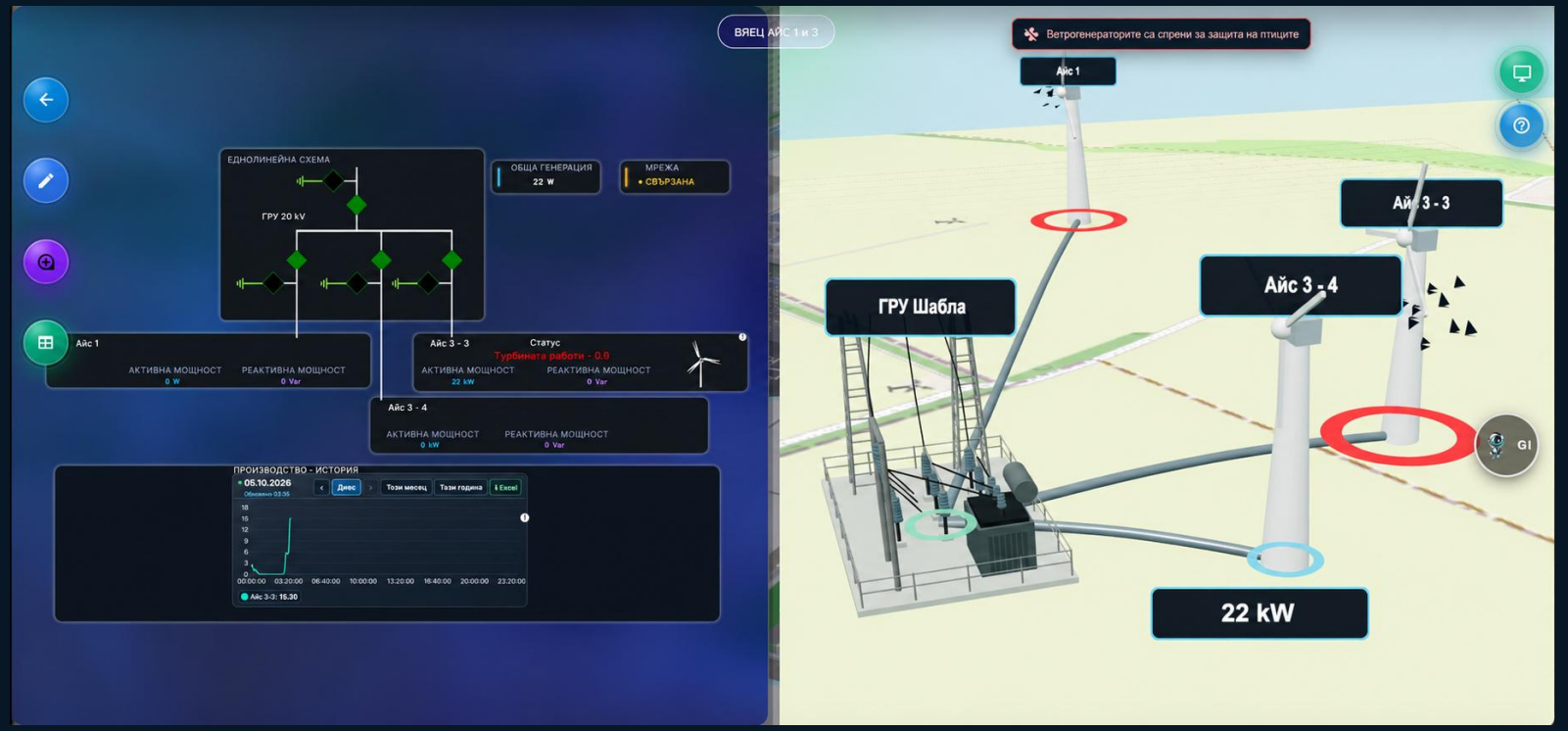
Forecast → actual → bias correction

Wind methodology now being extended

From Perception to Operational Action

Bird-migration risk can become a real-time operating constraint for wind generation.

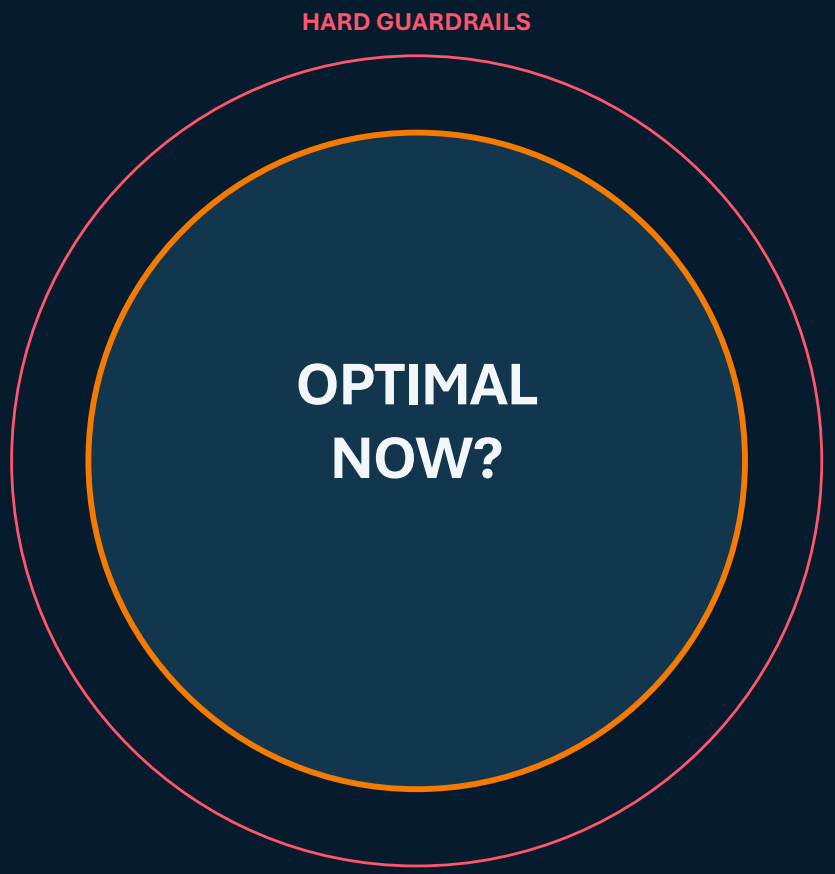
DETECT & TRACK | **ASSESS RISK** | **SLOW / STOP**



Environmental conditions become operational constraints — sensed, assessed and acted on under policy.

What Does “Optimal” Mean Right Now?

The best energy decision changes with business priorities — while hard constraints remain fixed.



ECONOMICS

cost • revenue • cash flow

ASSET LIFE

degradation • warranty • lifecycle

RELIABILITY

availability • reserve • continuity

ENVIRONMENT

CO_2 • ecological constraints

PRIORITY CAN CHANGE

Balanced • Cash-flow • Asset-life • Reliability • Revenue • Green

The objective can change. Safety, equipment, environmental and contractual guardrails do not.

Electricity Is Cheap. Should the Battery Charge Now?

At larger BESS scale, a price spread is an opportunity — not automatically a business decision.

MARKET OPPORTUNITY

13:00	18:00 forecast
€45 / MWh	→
	€100 / MWh

illustrative

NAÏVE ANSWER

CHARGE NOW

GRID-ONE DECISION

- **MARKET VALUE** — price spread • market schedule
- **ASSET HEALTH** — efficiency • degradation • SoC / SoH
- **PHYSICAL LIMITS** — production • reserve • grid / transformer
- **FUTURE VALUE** — next opportunity • reserve optionality

FULL • PARTIAL • HOLD • DISCHARGE • RESERVE



REAL GRID-ONE DEPLOYMENT

Belozem Solar Park 2 Ltd

36 MW / 72 MWh BESS • IBEX pricing used for scheduling

Doing nothing can be an optimized decision.

Belozem Solar Park 2 — One 36 MW / 72 MWh Virtual Battery

A complex physical installation presented to the customer as one controllable resource



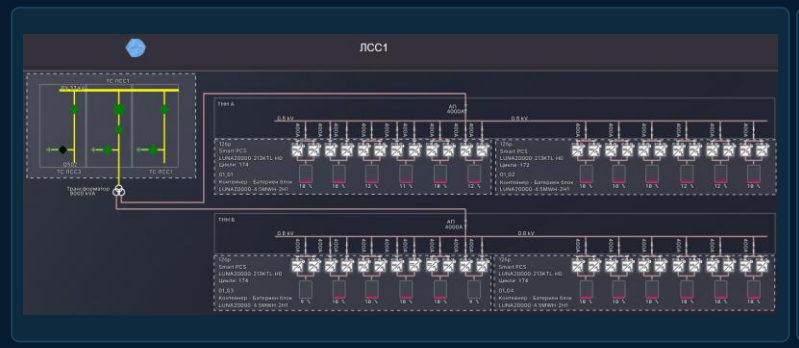
CUSTOMER VIEW
one resource

GRID-ONE VIEW
a managed hierarchy

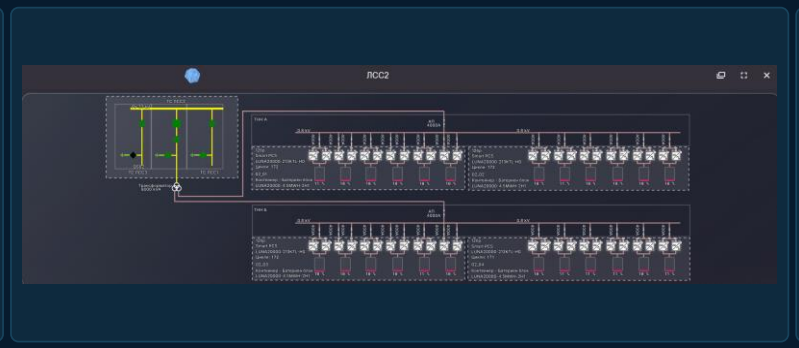
Belozem: Five Virtual BESS Under One Resource

LCCS1–LCCS5 are coordinated beneath the 36 MW / 72 MWh site-level virtual battery.

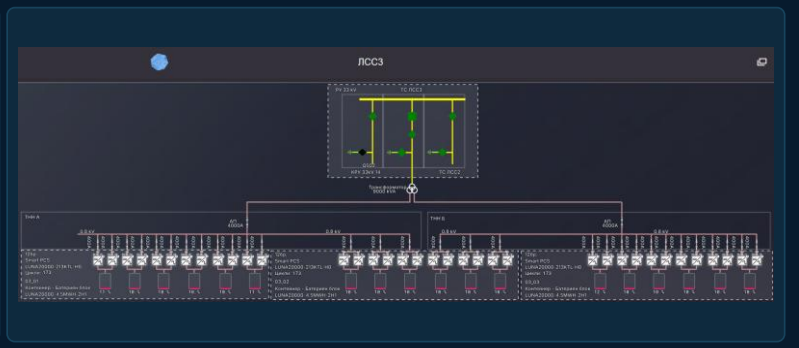
LCCS1



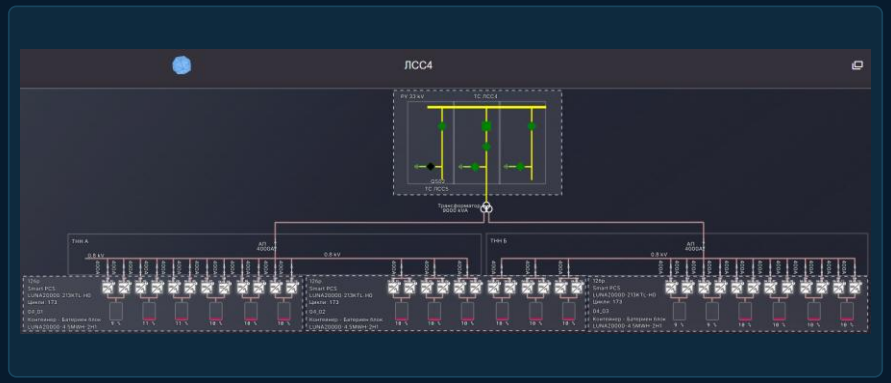
LCCS2



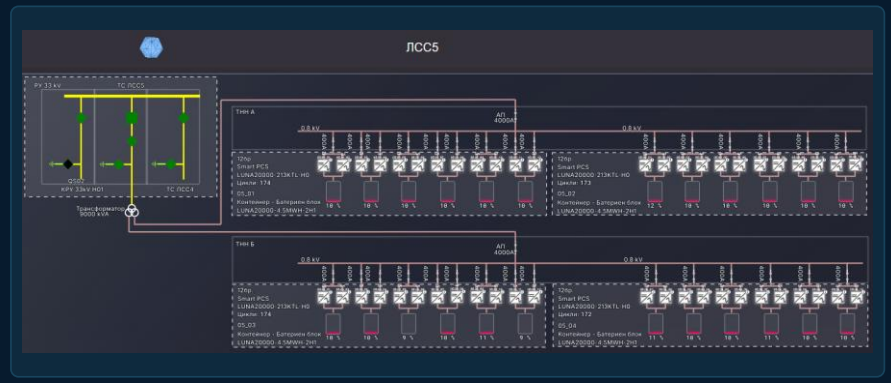
LCCS3



LCCS4



LCCS5



SITE → 5 VIRTUAL BESS → CONTAINERIZED VIRTUAL BESS → MANAGED BATTERIES

The customer sees one resource; Grid-One coordinates the hierarchy underneath.

AI Does Not Stop at the Control Command

Grid-One turns operational history into performance, business impact and recommendations.

CONTROL → MEASURE → EXPLAIN → RECOMMEND → RE-OPTIMIZE



MEASURE

energy • efficiency
alarms • tracking

EXPLAIN

cost / revenue
losses • root causes

RECOMMEND

optimization options
next actions • reports

Illustrative analytical workspace

Control closes the physical loop. Reporting closes the learning and business loop.

Grid-One Energy Community — Shared Flexibility, Coordinated Locally

Producers, consumers, storage and flexible loads can act together without losing local autonomy.

COMMUNITY ASSETS
PV • wind • BESS • EV
buildings • industry

LOCAL OBJECTIVES
comfort • production
availability • asset limits

GRID-ONE ROLE
forecast • schedule • optimize
allocate shared flexibility

EXTERNAL CONTEXT
market • ESO / grid
contracts • community rules



Local autonomy + shared coordination = usable flexibility for the community, grid and market.

What If Energy Could Think?

It would continuously decide how assets should work together — and explain the result.

SEE
physical state

PREDICT
what is likely

PLAN & SCHEDULE
what should happen

OPTIMIZE
across objectives

DECIDE
inside policies

ACT
through control

LEARN & REPORT
measure • explain • improve

The future energy system is not simply connected.
It continuously decides how its assets should work together.

Dynamic • Multi-objective • Lifecycle-aware • Resilient

Thank you



T&D ENGINEERING

Address: Bulgaria, Varna, str. Perunika 46

Phone: +052 312 100

E-mail: office@tid-engineering.bg

grid-one.bg • grid-one.eu